

Night Shift

A prize for the first machine that works through a lunar night.

Apollo did it in 1969 with fuel you can't buy.

Chang'e-4 does it today with fuel you can't buy, while the spacecraft sleeps.

Do it awake, at ten watts, with fuel you can order.

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Owner MoonDAO

Status Pre-registration. Nothing is on-chain and no rules are final.

Purpose Structured expert feedback before launch.

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The problem

A lunar night is **354 hours** of darkness at roughly **100 K**. Almost every spacecraft ever landed on the Moon dies in it, or sleeps through it.

The consequence is that lunar surface assets are **disposable**. A commercial lander is a two-week instrument that becomes a monument at sunset. A rover spends half its life unconscious. Nothing accumulates, every mission starts from zero, and the Moon acquires no installed base. A sustained human presence is not possible on hardware that has to be replaced every fortnight.

The technical problem is not warmth. It is **useful electrical work, continuously, without degrading, from a source you can actually buy**.

Why this isn't already solved

It looks solved, and the first reaction of most experts is that it is. Survival certainly is — Apollo settled that in 1969, and a 42 g radioisotope heater will keep a small box alive indefinitely. But there is a precise gap between what has been demonstrated and what a permanent asset requires, and it has two halves.

First half: nobody operates. They hibernate.

The strongest counter-example is Chang'e-4, which has been on the lunar far side since January 2019 and is the closest prior art in existence:

CHANG'E-4 LANDER

Radioisotope output	2.5 We (auxiliary), plus 120 W thermal
Spacecraft state at night	Sleep mode — the bus cannot run on it
What the power does	Runs a thermometer. Measured soil to −196 °C
The data	Stored, transmitted after dawn

Apollo's SNAP-27 did roughly **70 We** and ran for years. So the physics has been settled twice, decades apart. What has never been demonstrated is a machine that stays *awake and working* at a power level that constitutes work rather than instrumentation.

Second half: both proofs run on fuel nobody can purchase.

PLUTONIUM-238

US production	DOE targets 1.5 kg/yr; actual output below 0.5 kg/yr
Cost	> \$8M per kilogram
Availability	State reserve, allocated to flagship science missions
A single MMRTG	~\$54M, ~\$83M including launch approval
Can a company buy some?	No. Not at any price.

This is what the prize actually targets. It is an **industrialization** milestone, not a physics breakthrough — the difference between a capability two governments hold and a product anyone can buy. We would rather state that plainly than oversell it.

What clearing it unlocks

- **Mission duration.** Two-week commercial landers become month-class and beyond — the largest available multiplier on money already being spent.
- **Shadowed-crater operations.** Inside a permanently shadowed region there is no dawn to wait for. Ice prospecting is night operation by definition.
- **Continuous science.** A seismometer that sleeps half its life is half a seismometer. The same applies to heat flow, magnetometry, and dust monitoring.
- **An installed base.** The first hardware still working when the next lander arrives is the beginning of infrastructure rather than a sequence of expeditions.

What it does not unlock, stated plainly: nothing here scales to a habitat, a crew, or a resource-extraction plant. Those need kilowatts to megawatts and remain a fission problem. Ten watts is the difference between a dead lander and a working instrument station.

The prize

Deliver at least 10 watts of electricity, continuously, for 354 hours, in a vacuum chamber colder than 100 kelvin, with nothing plugged in — and still be delivering the same 10 watts at the end. Publish the data.

It is a terrestrial thermal-vacuum test, and there is **one milestone**. You either cleared it or you didn't.

PARAMETER	REQUIREMENT
Delivered power	≥ 10.0 We, every rolling 60-minute window
Duration	354 continuous hours
Radiative sink	≤ 100 K (liquid nitrogen at 77 K qualifies)
Pressure	≤ 1×10 ⁻⁵ torr
Energy source	Entirely onboard at T0
Output retention	Final hour ≥ 98% of first hour
Purse	\$25,000, growing with betting volume
Deadline	None — open until solved

Three rules carry the whole design

1 · THE UNPLUG RULE

Nothing crosses the chamber boundary into the article except data.

No power, heat, propellant, coolant, or mass. One rule applied identically to every chemistry — no technology ban and no carve-out for nuclear. A wall-powered heater standing in for a radioisotope fails, because the chamber wall would be the power source. The same heater running off an onboard battery passes; it simply spends the entrant's own energy.

2 · THE SUSTAINED OUTPUT RULE

Power in the final hour must be at least 98% of power in the first hour.

This is how the prize stays technology-agnostic while remaining grounded in what a permanent asset needs. An output that decays across one night is a consumable with a fortnight of runway — the problem we are trying to retire. Any isotope passes without trying: americium-241 decays 0.007% over 354 hours. Any consumable must be oversized until 354 hours is a small bite out of its inventory, which costs mass, which the public scoreboard then reports.

3 · THE PROCUREMENT RULE

The energy source must be commercially procurable, evidenced by an invoice or a published price.

Not a technology ban — it says nothing about which physics you use, only that the answer must be one somebody else could buy. State-reserve plutonium fails because it is a government furnishing and no invoice exists. Americium-241 passes at roughly \$1,500/g, with commercial offtake agreements signed in 2025. Batteries and chemical systems pass off the shelf. A useful side effect is that nationality never enters the rules.

How it is judged

Nobody judges. A script does. Everything that decides the outcome is a number in a data file, so we publish an open-source validator and any bettor, rival, or journalist can run it on the same files and get the same answer:

VALIDATOR CHECK	THRESHOLD
Minimum rolling 60-minute mean power	$\geq 10.0\text{ W}$
Minimum instantaneous power	$\geq 5.0\text{ W}$
Longest interval below floor	$\leq 60\text{ s}$
Final-hour $\div$ first-hour mean power	≥ 0.98
Contiguous duration	$\geq 354\text{ h}$
Maximum shroud temperature, all sensors	$\leq 100\text{ K}$
Maximum chamber pressure	$\leq 1 \times 10^{-5}\text{ torr}$

That only works if the files are trustworthy, so a second rule governs custody: **an entrant may own the chamber and the article, but never the instruments or the data.** Whoever calibrates, installs, seals, and reads the measurement chain must be independent of the entrant.

Crucially, **observation is not judgment.** The verifier decides nothing about whether an entry counts and needs no opinion about lunar power systems. They attest to facts a competent person can observe: what the article weighed, which conductors were in the feedthrough panel, whose calibrated sensors these were, and that the seals held. Verification is two attended days — sealing on day zero, custody of the files on day fifteen — with tamper-evident seals and continuous video covering the gap.

The field

Competitors are named by MoonDAO from public sources and do not apply — they claim. The roster follows one selection rule: **complete systems, not hosts.** If a lander flies somebody else's power box, that is one result and it belongs to the box.

ORGANIZATION	APPROACH	POSITION AS OF SEPT 2026	READ
Zeno Power United States	Am-241 and Sr-90 radioisotope, Stirling conversion	\$15M NASA "Harmonia" Space Act Agreement. Final design review April 2026 at 3.5× the original power spec. Orano americium offtake signed Sept 2025. Terrestrial demo due early 2027	The favourite. Retention rule is nearly free for an isotope
Astrobotic (NITE) United States	Chemical / fuel cell — non-nuclear	~\$2.8M NASA Tipping Point award. Thermal-vacuum testing at Penn State hit leakage and an oxidizer pump failure. Roughly TRL 4	Only funded non-nuclear operate-through system
Venturi Space Monaco / Switzerland / France	Cryogenic battery packs	Cells qualified −240 °C to +130 °C. Lunar-cycle vacuum endurance testing already inside their qualification flow. Supplies Astrolab's NASA-selected crewed rover	Could clear this fastest of anyone
Perpetual Atomics + Univ. of Leicester United Kingdom	Am-241 heater units, ESA ENDURE programme	First americium-fuelled heater core produced April 2026 with QSA Global. Vibration qualified; thermal-vacuum only to −70 °C so far	Needs a converter and a genuinely cryogenic run
CNNC / China Institute of Atomic Energy China	Pu-238 radioisotope generators	Only flight-proven lunar night electricity outside Apollo. National RTG standard GB/T 44319-2024 in force since March 2025	The prior art. State fuel fails the Procurement Rule
Rosatom / Kurchatov Institute Russia	Radioisotope heritage; "Selena" surface reactor	Deep Lunokhod-era heritage. Current programme is a 5–10 kWe reactor, prototype 2032, deployment ~2036	Wrong scale and wrong decade for this bar
ISRO / BARC India	Am-241 heater units	Heater flown as a demonstrator on Chandrayaan-3's propulsion module. Targeting 100–200 day lander life; LUPEX is the named night demonstrator	Aimed at heat, not watts
Open Field	Any qualifying entrant not listed	University labs, unannounced startups, national labs, small teams	Genuine, not decorative

Deliberately excluded as integrators or customers rather than competitors: Firefly Aerospace, Intuitive Machines, Blue Origin, SpaceX, and ispace. Excluded on scale: tritium and betavoltaic cells, which are three to four orders of magnitude below the bar.

How the prize itself works

Night Shift runs on MoonDAO's DePrize mechanism, which pairs a prize with a prediction market. Each named competitor is a tradeable outcome; participants back who they think will get there first; live odds emerge from the trading. A share of betting volume accrues to the prize pool, so the purse grows as interest does.

Two design consequences are worth flagging to a reviewer. First, **the prize has no deadline** — it stays open until somebody clears it. Second, and more importantly, the prize does not need to be entered to do its job. Alongside the market we publish a standing **scoreboard** of every listed competitor's best publicly verified result against this exact standard. Most cells will start empty. That table is the point: it gives the market something to price every month rather than one binary event years away, and an empty cell beside a rival's filled one is an argument inside somebody's own building. Getting organizations to disclose more about where they actually stand is an explicit objective, not a side effect.

What we are asking you for

This brief is a request for criticism, not endorsement. The prize is unregistered and nothing is on-chain, so every number below is still movable. We are specifically *not* looking for a view on whether lunar night power matters — we are looking for the flaw that makes this prize the wrong instrument, or the rule that a competent engineer would walk straight through.

The questions that would most change the design

- 1. Can you beat the validator?** Describe a data file that passes every check on page 3 alongside hardware that shouldn't count.
This is the single most valuable thing a reviewer can give us.
- 2. Is 10 watts the right bar?** It is four times Chang'e-4's entire nighttime budget, and it implies roughly \$1.4M of americium for an isotope entry. Correct, or should it be five?
- 3. Is 98% retention measuring the right thing?** It is trivial for any isotope and hard for any consumable, which is the intent.
Too tight, too loose, or the wrong property altogether?
- 4. Is two attended days enough independence?** Seals and video cover the 350 hours in between. Can that be defeated?
- 5. Would a verification body actually take this on?** The design assumes a firm of the DNV or TÜV type will witness a fifteen-day cryogenic vacuum test for a few thousand dollars. Is that realistic?
- 6. Who runs a chamber for fifteen continuous days?** This is our largest unpriced risk and the practical gate on how large the field gets. Named suggestions are more useful than anything else in this list.
- 7. Who have we missed?** Particularly outside the United States and Europe, and particularly non-nuclear approaches.
- 8. Is the framing too modest?** We claim an industrialization milestone rather than a physics breakthrough. Is that worth a prize at all?

What happens with your feedback

Responses go into a revision of the full rules document, which runs to considerably more detail than this brief — definitions, excursion and restart handling, publication requirements, safety, the verification routes, and the market mechanics. We are happy to share it with anyone who wants to go deeper, and equally happy to take fifteen minutes of objections over a written response.

Rules freeze when the market opens, and after that only clarifications that cannot change an outcome are permitted. So this window — before registration — is the only one in which fundamental changes are cheap.

The one-sentence version, if you only remember one thing: we are trying to move lunar night power from a capability two governments have demonstrated with unobtainable fuel to a product with a price and a lead time — and to make the industry show its working while it happens.

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Confidential, circulated under NDA. Competitor listings are editorial and drawn from public sources; listing does not imply participation, endorsement, or affiliation. Figures for plutonium-238 production and cost are from US Department of Energy and Government Accountability Office reporting; Chang'e-4 power figures from published mission literature and China Academy of Space Technology statements. Full source list available in the complete rules document.